

Patrick Mayer – Game Developer

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SKILLS

- **Primary Software/Languages:** Godot, Unity, C++, C, C#, Git, CMake
- **Secondary Software/Languages:** WebGPU, WebGL, Makefile, HTML/CSS, BASH, Java
- **Certifications:** FDI Design Certification of Excellence

EXPERIENCE

Florida Poly (2025 through 2026) | *Game Development Researcher*

- Designed and implemented game features such as minecart physics, a basic water shader, and boss fight scripts.

PROJECTS

Hive Mind Nebula 2 (Fall 2025) | WebGPU, JS

- 6 DoF movement using quaternions with space momentum physics.
- Optimized projectile spawning and despawning using object pooling.
- Parsed and rendered 3D models in .obj and .mtl formats.
- Created boss attacks patterns with state machine.

Tower Takeover (Spring 2025) | Unity

- Multiplayer race-to-the-top 2D platformer with unique player abilities.
- Implemented platformer physics and items.
- Handled message passing and variable synchronization.

Lasso Up, Partner (Fall 2024) | Unity

- Implemented movement state machine, game manager/level manager, lasso physics, save/load, dynamic crosshair, and speedrun timer functionality.
- Created a class diagram for all significant classes.
- Learned how to collaborate on a large Unity codebase with git.

Captain Zip (Spring 2024) | Godot, GDScript

- Implemented collectable coins, sound effects, enemy shooting, and save/load functionality.
- Created the final level which rewards skill while still being accessible to beginners.
- Won 1st place at the 2024 Spring FL Poly Game Jam.

EXTRACURRICULAR ACTIVITIES

FL Poly Media Club

- Participated in a game jam every semester I was in college.
- Became treasurer my sophomore year.

EDUCATION

Florida Polytechnic University, 2021-2025, Lakeland, FL

Bachelor of Science in Computer Science

Relevant Coursework:

- Advanced Computer Graphics, Advanced Computer Vision, Game Design 1 & 2, Virtual Reality, Computer Graphics, OOP, Operating Systems, Intro to Unix, Advanced Algorithm Design and Analysis, Data Structures, Computer Animation, Database 1, Data Wrangling, Intro to Computer Networks, Discrete Math 1 & 2, Comp Linear Algebra, Computer Architecture, Secure Software Engineering, Physics 1